

II. FOURIER TRANSFORM

The Fourier transform is probably the most widely applied signal processing tool in science and engineering. It reveals the frequency composition of a time series $x(t)$ by transforming it from the time domain into the frequency domain. In 1807, the French mathematician Joseph Fourier found that any periodic signal can be presented by a weighted sum of a series of sine and cosine functions. However, because of the uncompromising objections from some of his contemporaries such as J. L. Lagrange (Herivel 1975), his paper on this finding never got published, until some 15 years later, when Fourier wrote his own book, The Analytical Theory of Heat (Fourier 1822). In that book, Fourier extended his finding to periodic signals, stating that an periodic signal can be represented by a weighted integral of a series of sine and cosine functions [5]. Such an integral is termed the Fourier transform. Using the notation of inner product, the Fourier transform of a signal can be expressed as

$$X(f) = \langle x, e^{i2\pi ft} \rangle = \int_{-\infty}^{\infty} x(t) e^{i2\pi ft} dt \quad (1)$$

Assuming that the signal has finite energy,

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty \quad (2)$$

Accordingly, the inverse Fourier transform of the signal $x(t)$ can be expressed as

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{-i2\pi ft} df \quad (3)$$

Signals obtained experimentally through a data acquisition system are generally sampled at discrete time intervals ΔT , instead of continuously, within a total measurement time T . Such a signal, defined as x_k , can be transformed into the frequency domain by using the discrete Fourier transform (DFT), defined as

$$DFT(f_n) = \frac{1}{N} \sum_{k=0}^{N-1} x_k e^{-i2\pi f_n k \Delta T} \quad (4)$$

Every signal can be written as a sum of sinusoids with different amplitudes and frequencies.

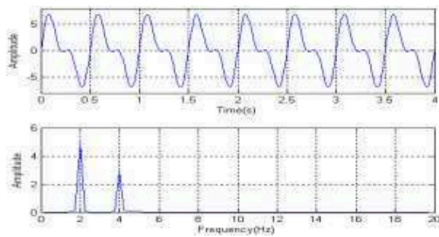


Fig.1: Discrete Fourier transforms to transform data from time into the frequency domain

→ comparison on Fourier and Wavelet Transformation Mohammad Heman Milon / p.2

① Fourier Transf.

A1-1 Fourier Transform in L^1 9

Fourier Transform

We can now give the basic definition.

DEFINITION A1.1. Let $s(t)$ be a stable complex signal. The Fourier transform (FT) of $s(t)$ is the function from $\mathbb{R}$ into $\mathbb{C}$:

$$\hat{s}(v) = \int_{\mathbb{R}} s(t) e^{-2i\pi vt} dt. \quad (1)$$

(Note that the argument of the exponential in the integrand is $-2i\pi vt$.) The mapping from the function to its Fourier transform will be denoted by

$$s(t) \xrightarrow{FT} \hat{s}(v) \text{ or } \mathcal{F}: s(t) \rightarrow \hat{s}(v).$$

Riemann-Lebesgue Lemma

The Riemann-Lebesgue lemma¹ is one of the most important technical tools in Fourier analysis, and we shall use it several times, especially in the study of pointwise convergence of Fourier series (Chapter A3).

THEOREM A1.2. The FT of a stable complex signal $s(t)$ satisfies

$$\lim_{|v| \rightarrow \infty} |\hat{s}(v)| = 0.$$

Breiman p.15

Im unendlichen verschwindet FT

aber Rauschen verschwindet nicht im ∞ Zeit konstant erhalten
→ Problem des FT mit verrauschten Signalen

→ Mathematical principles of Signal processing - Fourier and Wavelet Analysis - Breiman / p.9

? hier noch Konvergenzresultat?

② Inverse Fourier Transf.

A1-2 Inversion Formula 17

THEOREM A1.4. Let $s(t)$ be an integrable complex signal with the Fourier transform $\hat{s}(v)$. Under the additional condition

$$\int_{\mathbb{R}} |\hat{s}(v)| dv < \infty, \quad (11)$$

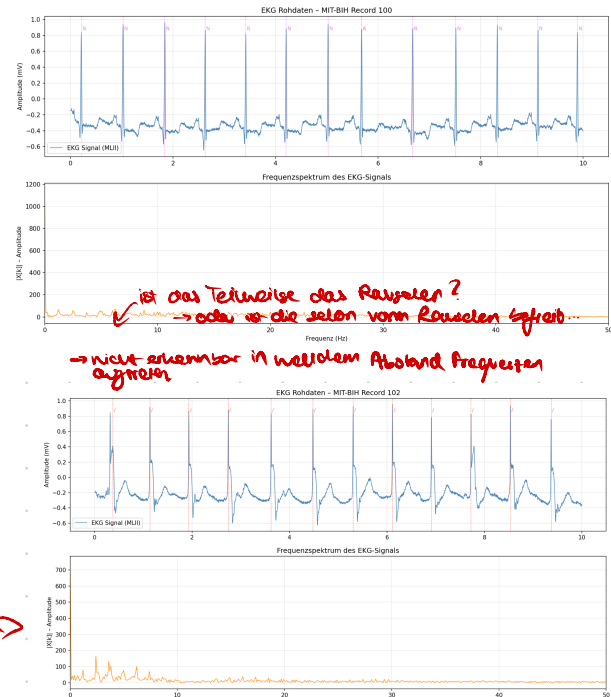
the inversion formula

$$s(t) = \int_{\mathbb{R}} \hat{s}(v) e^{+2i\pi vt} dv \quad (12)$$

holds for almost all t . If $s(t)$ is, in addition to the above assumptions, continuous, equality in (12) holds for all t .

(Note that the exponent of the exponential of the integrand is $+2i\pi vt$.)

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+ immer keine Teilkoerung
→ Milon / p.4

③ Einführung zu Verarbeitung diskreter Daten:

The Weak Poisson Formula

The Poisson sum formula takes many forms. The strong version is

$$T \sum_{n \in \mathbb{Z}} s(nT) = \sum_{n \in \mathbb{Z}} \hat{s}\left(\frac{n}{T}\right). \quad (30)$$

This aesthetic formula has a number of applications in signal processing (see Part B).
→ Basis für FFT

The next result establishes the connection between the **Fourier transform** and Fourier series, and is central to sampling theory. It is a weak form of the Poisson sum formula (see the discussion after the statement of the theorem).

THEOREM A2.3. Let $s(t)$ be a stable complex signal, and let $0 < T < \infty$ be fixed. The series $\sum_{n \in \mathbb{Z}} s(t + nT)$ converges absolutely almost everywhere to a T -periodic locally integrable function $\Phi(t)$, the n th Fourier coefficient of which is $(1/T)\hat{s}(n/T)$.

We paraphrase this result as follows: Under the above conditions, the function

$$\Phi(t) := \sum_{n \in \mathbb{Z}} s(t + nT) \quad (31)$$

is T -periodic and locally integrable, and its formal Fourier series is

$$S_f(t) = \frac{1}{T} \sum_{n \in \mathbb{Z}} \hat{s}\left(\frac{n}{T}\right) e^{2i\pi \frac{n}{T} t}. \quad (32)$$

(We speak of a "formal" Fourier series, because nothing is said about its convergence.) Therefore, whenever we are able to show that the Fourier series represents the function at $t = 0$, that is, if $\Phi(0) = S_f(0)$, then we obtain the Poisson sum formula (30).

For now, we are saying nothing about the convergence of the Fourier series. This is why we talk about a *weak* Poisson's formula. A *strong* Poisson's formula corresponds to the case where one can prove the equality *everywhere* (and in particular at $t = 0$) of $\Phi(t)$ and of its Fourier series. We shall say more about the Poisson formula and, in particular, give strong versions of it in Section A3.3. The version we have here, and that we shall proceed to prove, is the one we need in the Shannon-Nyquist sampling theorem (Chapter B2).

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④ DFT

B3.1 The DFT and the FFT Algorithm

The DFT

Suppose we need to compute numerically the FT of a stable signal $s(t)$. In practice only a finite vector of samples is available,

$$s = (s_0, \dots, s_{N-1}), \rightarrow \text{unser Signaldaten}$$

where $s_n = s(n\Delta)$. The Fourier sum of this vector evaluated at pulsations $\omega_k = 2k\pi/N$ is the **discrete** Fourier transform (DFT).

DEFINITION B3.1. The DFT of $s = (s_0, \dots, s_{N-1})$ is the vector $S = (S_0, \dots, S_{N-1})$, where

$$S_k = \sum_{n=0}^{N-1} s_n e^{-i(2\pi kn/N)}.$$

The DFT is an approximation of the FT, the quality of which depends on the parameters N and Δ . The first question to ask is: How to choose these parameters to attain a given precision? As we shall see, the answer is given by the Poisson sum formula. For the time being, we shall give the basic properties of the DFT without reference to a sampled signal.

Let $a = (a_0, \dots, a_{N-1})$ be a finite sequence of complex numbers. For the N th root of unity, we adopt the following notation:

$$w_N = e^{-i\frac{2\pi}{N}}.$$

The finite sequence $A = (A_0, \dots, A_{N-1})$ defined by

$$A_m = \sum_{n=0}^{N-1} a_n w_N^{mn} \quad (89)$$

is the DFT of $a = (a_0, \dots, a_{N-1})$.

THEOREM B3.1. We have the inversion formula

$$a_n = \frac{1}{N} \sum_{m=0}^{N-1} A_m w_N^{-mn}. \quad (90)$$

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FFT

The Fast Fourier Transform Algorithm

The calculation of the DFT of the sequence $\{a_n\}$ by formula

$$A_m = a_0 + a_1 w_N^m + a_2 w_N^{2m} + \dots + a_{N-1} w_N^{m(N-1)}$$

requires $N - 1$ multiplications for $m \geq 1$ (none for $m = 0$). If we consider that the cost of an addition is negligible compared with that of a multiplication, the calculation of $A = (A_0, \dots, A_{N-1})$ thus requires $(N - 1)^2$ computational units,

where one unit corresponds to one multiplication. The fast Fourier algorithm,⁵ also called the fast Fourier transform (FFT), considerably reduces the computational complexity. It is based on the following remark.

Let $a_n \leftrightarrow A_m$ be a DFT pair (note that we are considering a DFT of order $2N$ with $2N$ terms a_n and $2N$ terms A_m). Define

$$b_n = a_{2n}, \quad c_n = a_{2n+1} \quad (0 \leq n \leq N-1),$$

and

$$b_n \leftrightarrow_N B_m, \quad c_n \leftrightarrow_N C_m$$

(the latter DFTs are of order N). A direct calculation shows that

$$A_m = B_m + w_{2N}^m C_m \quad (0 \leq m \leq 2N-1). \quad (99)$$

Observing that $B_{m+N} = B_m$, $C_{m+N} = C_m$, and $w_{2N}^{m+N} = -w_{2N}^m$, we can split Eq. (99) in two parts:

$$A_m = B_m + w_{2N}^m C_m \quad (0 \leq m \leq N-1) \quad (100)$$

and

$$A_{m+N} = B_m - w_{2N}^m C_m \quad (0 \leq m \leq N-1). \quad (101)$$

In order to calculate B_m and C_m for $0 \leq m \leq N-1$, we need $2(N-1)^2$ computational units. When (100) is used we need $N-1$ additional multiplications. The multiplications in (101) are for free since they were done in (100). In total, the method requires

$$2(N-1)^2 + N - 1 = (N-1)(2N-3)$$

units instead of $(2N-1)^2$ for the direct method. If we have to calculate a DFT of order N such that

$$N = 2^r, \quad (102)$$

the FFT will take $F(N) \leq \frac{1}{2} N \log_2 N$ computational units. The result is obtained by induction. Indeed, $F(2) = 1$, and the considerations above show that

$$F(2N) = 2F(N) + N - 1 \leq 2F(N) + N.$$

But if $F(N) \leq \frac{1}{2} N \log_2 N$, then $2F(N) + N \leq N(\log_2 N + 1) = \frac{1}{2} 2N \log_2 2N$.

→ hab jetzt noch nicht ganz gesehen was FFT von DFT unterscheidet...

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Jetzt: Theorie hinter De-noising - erste Lieder in extra Kapitel?

5.2 Thresholding Methods

These methods use a threshold and determine the clean wavelet coefficients based on this threshold. There are two main ways of thresholding the wavelet coefficients, namely

- Soft-Thresholding Method.
- Hard-Thresholding Method.

5.2.1 Hard Thresholding Methods

If the absolute value of a coefficient is less than a threshold, then it is assumed to be 0, otherwise it is unchanged. Mathematically it is

$$X = \text{sign}(Y)(Y \cdot (\text{abs}(Y) - \lambda))$$

where Y represents the noisy coefficients, λ is the threshold, X represents the estimated coefficients.

5.2.2 Soft Thresholding Methods

Hard thresholding is discontinuous. This causes ringing/Gibbs effect in the de-noised image. To overcome this, Donoho introduced the soft thresholding method.

If the absolute value of a coefficient is less than a threshold λ , then it is assumed to be 0, otherwise its value is shrunk by λ . Mathematically it is

$$X = \text{sign}(Y)(Y \cdot ((\text{abs}(Y) - \lambda) \cdot \text{abs}(Y) - \lambda))$$

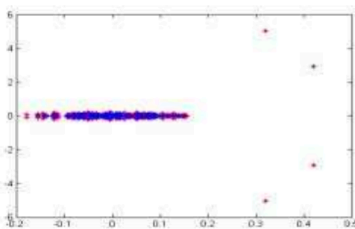


Fig.4(a): Noisy Signal

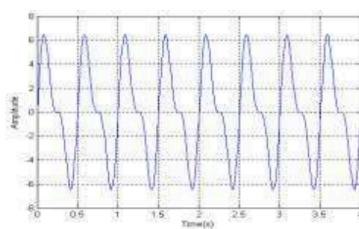


Fig.4 (b): Signal de-noising with the help of Fourier transformation

— Milon p. 5

→ dafür noch mathematische Grundlage finden...

→ setzt voraus das Rauschen im Zeitbereich gleichmäßig verteilt ist weil FT gleichmäßig verteilt
→ Problem: Rauschen ist nicht unbedingt gm verteilt...
→ wieder keine Zeitlokalisierung